

Experiment 1: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1988	2.273	1.258	4.946	38.7%	73.2%	84.7%	14.423	27.808	9.573	26.347	0.561	1.293	1.434
Proposed	1981	2.359	1.028	3.971	48.9%	79.0%	87.0%	15.492	29.665	7.521	16.499	0.536	1.284	1.501
PPO-GRU	1991	2.890	1.298	9.229	36.5%	71.2%	83.5%	17.617	40.716	8.010	17.477	0.528	1.357	1.510
MPC	1053	5.891	3.049	15.983	2.8%	17.6%	47.6%	3.013	3.704	1.582	2.822	0.166	0.148	0.208

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1988	99.4%	12	0.6%	0.6%	12	100.0%	0	0
Proposed	2000	1981	99.0%	19	0.9%	0.9%	19	100.0%	0	0
PPO-GRU	2000	1991	99.5%	9	0.5%	0.5%	9	100.0%	0	0
MPC	2000	1053	52.6%	947	47.4%	47.4%	947	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	947	936	98.8%	198	21.2%	2.574
Proposed	947	934	98.6%	142	15.2%	2.560
PPO-GRU	947	941	99.4%	155	16.5%	3.005